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INTERNSHIP TITLE : AI PROCTORING SYSTEM

COMPANY NAME : MAANTT GLOBAL SERVICES PRIVATE LIMITED

Software company in Chennai, Tamil Nadu

TABLE OF CONTENTS

S.NO	SECTIONS	PAGE NO
1	INTRODUCTION	1
2	MY ROLE	1
3	PURPOSE OF THIS REPORT	1
4	ACTIVITIES	1
5	RESPONSIBILITIES	1
6	CHALLENGES	1-2
7	SKILLS LEARNED	2
8	CONTRIBUTIONS	3
9	CONCLUSION	3
10	MY PROJECTS	4

INTRODUCTION

MAANTT is a distinctive company that provides platform-based solutions tailored to corporate talent development requirements and technology-driven learning solutions for educational institutions. The company is committed to delivering digital solutions in the retail and life care sectors to enhance user experience through process optimization. Additionally, MAANTT empowers young graduates by engaging them in innovative projects designed to inspire the global community.

MY ROLE

I worked as a team member in building an AI proctoring system.

PURPOSE OF THIS REPORT

I have attached my internship work report for your reference, which summarizes my responsibilities, achievements, and key learnings during the internship period.

ACTIVITIES

1. Completed two key tasks in the proctoring requirements worksheet:
 - **Candidate Authorization** (verifying candidate identity)
 - **Auto Authorization** (automating approval processes)
2. **Integration:** Merged these two tasks into a single streamlined workflow.
3. **Collaboration:** Combined the integrated solution with other team members' modules.
4. **API Integration:** Assisted in connecting the final solution to a **bidirectional WebRTC API** for real-time proctoring communication.

RESPONSIBILITIES

1. **Designed and implemented a MySQL dataset** to accurately demonstrate the Auto Authorization and Candidate Authorization processes, making necessary adjustments to ensure proper functionality.
2. **Took ownership of integrating my work** with other team members' components, ensuring seamless system interoperability as we developed the proctoring solution.
3. **Played a key role in the API implementation phase**, where I researched, identified, and successfully connected our integrated system to a bidirectional WebRTC API to enable real-time communication capabilities.

CHALLENGES

1. **Compatibility Issues:**
 - Ensuring seamless integration between **Candidate Authorization** and **Auto Authorization** modules, as they may have had different data formats or dependencies.
2. **Real-Time Data Synchronization:**
 - Managing delays or conflicts when merging multiple modules before API connectivity.

3. **Bidirectional WebRTC Integration:**

- Handling **peer-to-peer connectivity issues**, such as NAT traversal, firewall restrictions, or unstable connections.
- Ensuring **low-latency, high-quality streaming** for proctoring without interruptions.

4. **Security & Privacy Concerns:**

- Implementing secure authentication while maintaining smooth auto-approval processes.
- Preventing unauthorized access in the WebRTC communication channel.

5. **Error Handling & Debugging:**

- Resolving conflicts when combining different team members' code into a unified system.
- Logging and troubleshooting API failures or authorization mismatches.

SKILLS LEARNED

Technical Skills:

1. **Database Management** - Designed and implemented MySQL datasets for authorization systems
2. **System Integration** - Successfully merged independent modules into a unified workflow
3. **API Development** - Researched, selected and integrated WebRTC bidirectional APIs
4. **Real-time Communication Systems** - Implemented WebRTC for low-latency data streaming
5. **Authorization Protocols** - Developed both automated and manual candidate verification processes
6. **Debugging & Troubleshooting** - Resolved integration challenges between system components

Professional Skills:

1. **Collaborative Development** - Worked effectively in a team integration environment
2. **Technical Documentation** - Created datasets and system demonstrations
3. **Solution Research** - Evaluated and selected appropriate technical solutions (WebRTC)
4. **Ownership & Accountability** - Took responsibility for critical integration tasks
5. **Problem-Solving** - Addressed compatibility and functionality challenges
6. **Technical Communication** - Explained and demonstrated system functionality

The Python libraries that I had learned are

- pandas
- numpy
- matplotlib
- seaborn
- dlib
- face_recognition

- flask
- opencv
- websocket

CONTRIBUTION

1. Development & Implementation

- **Designed and configured a MySQL database** to simulate real-world testing scenarios, enabling accurate validation of:
 - **Auto Authorization** (automated candidate verification)
 - **Candidate Authorization** (manual identity checks)
- **Refined authorization logic** by adjusting validation rules, ensuring seamless authentication workflows.

2. System Integration & Collaboration

- **Took ownership of merging my modules** (Auto + Candidate Auth) with other team members' components, ensuring smooth interoperability.
- **Resolved compatibility issues** between different system segments, maintaining data consistency and process efficiency.

3. API & Real-Time Connectivity

- **Researched and identified WebRTC** as the optimal solution for bidirectional proctoring communication.
- **Successfully integrated WebRTC API** into the system, enabling real-time video/audio monitoring for remote proctoring.

4. Problem-Solving & Quality Assurance

- **Debugged integration challenges**, including data flow mismatches and API connectivity issues.
- **Ensured secure and low-latency performance** in the WebRTC implementation for reliable exam monitoring.

CONCLUSION

My internship as a Team Member at MAANTT Global Services Private Limited was an enriching professional experience where I contributed to developing an AI Proctoring System while significantly enhancing my technical and project management skills. Through hands-on involvement, I gained practical expertise in Python programming, library implementation, and agile task management within a live project environment. I successfully implemented key system components including candidate authorization modules and WebRTC integration, while collaborating effectively with cross-functional teams. I am deeply grateful to my mentor for their invaluable guidance and support throughout this transformative learning journey. This internship has equipped me with robust technical capabilities and real-world project experience that I look forward to applying in my future career endeavours in AI and software development.

MY PROJECTS

USECASE ANALYSIS

PROCESS

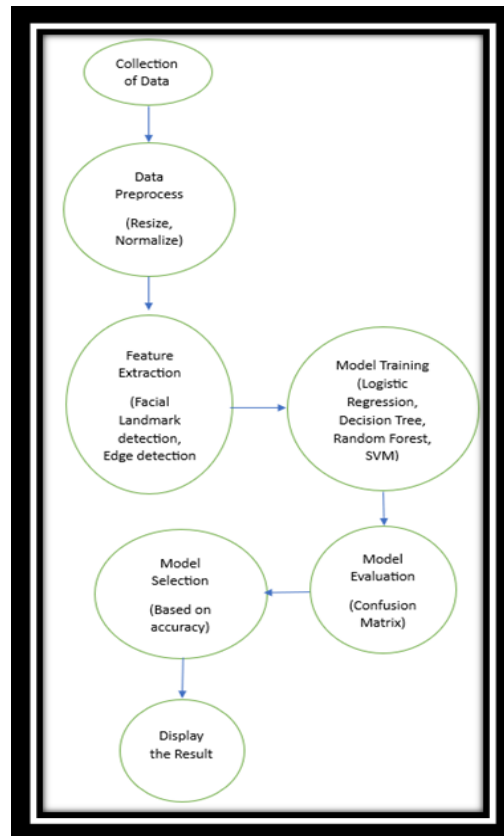


Figure 1: use case process

OUTPUT



Figure 2: candidate authorization output

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_6356\463978825.py:33: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connect
candidates = pd.read_sql(candidate_query, cnx)
C:\Users\Admin\AppData\Local\Temp\ipykernel_6356\463978825.py:34: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connect
proctors = pd.read_sql(proctors_query, cnx)
Proctor allocated to candidate
Match Found
Match Found
Match Found
Match Found
Match Found
Match Found
Match Found
Match Found
Match Found
Match Found
Match Found
Match Found
Checking again in 0.32 minutes
Checking again in 0.31 minutes
Checking again in 0.30 minutes
Checking again in 0.28 minutes
Checking again in 0.27 minutes
Checking again in 0.26 minutes
```

Figure 3: Auto authorization output

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_44656\463978825.py:33: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connectable
candidates = pd.read_sql(candidate_query, cnx)
C:\Users\Admin\AppData\Local\Temp\ipykernel_44656\463978825.py:34: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connectable
proctors = pd.read_sql(proctors_query, cnx)
Proctor allocated to candidate
Match Found
```

Figure 4: output for match not found

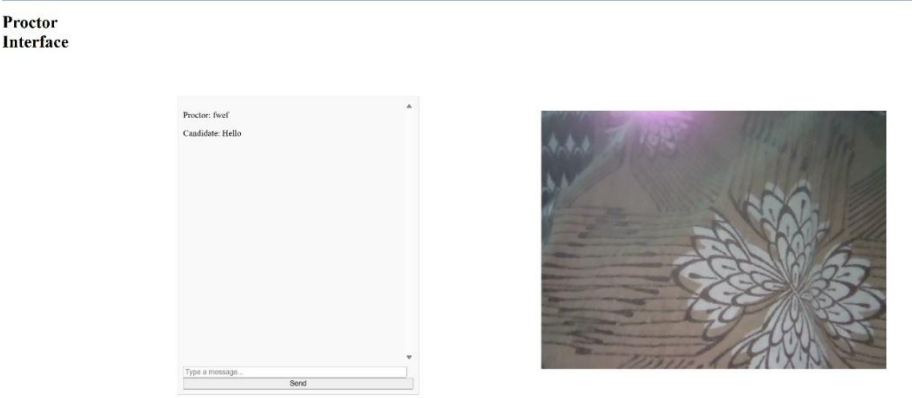


Figure 5: WebRTC connection (proctor interface)

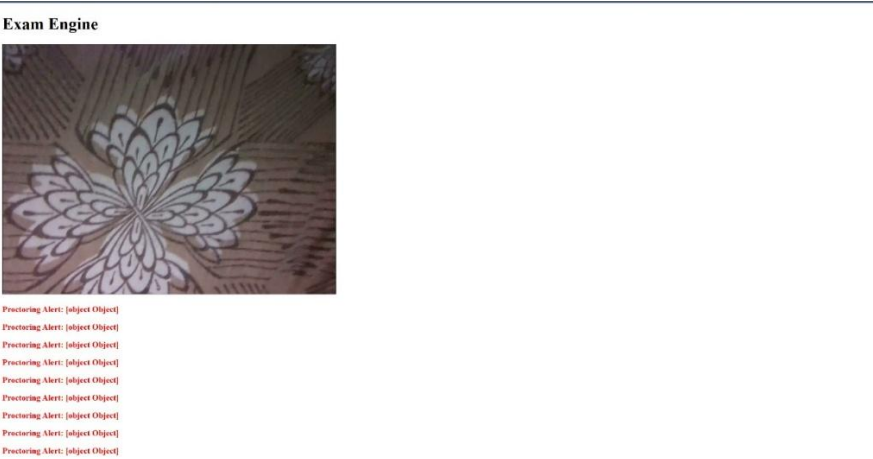


Figure 6: WebRTC connection (Exam Engine)

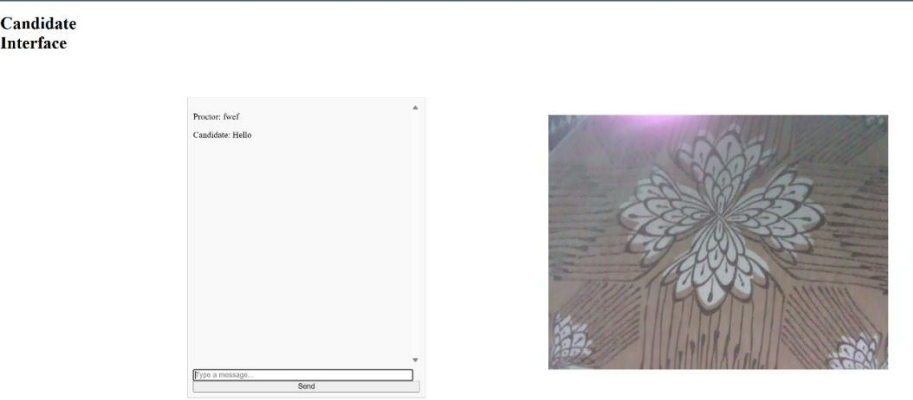


Figure 7: WebRTC connection (candidate interface)